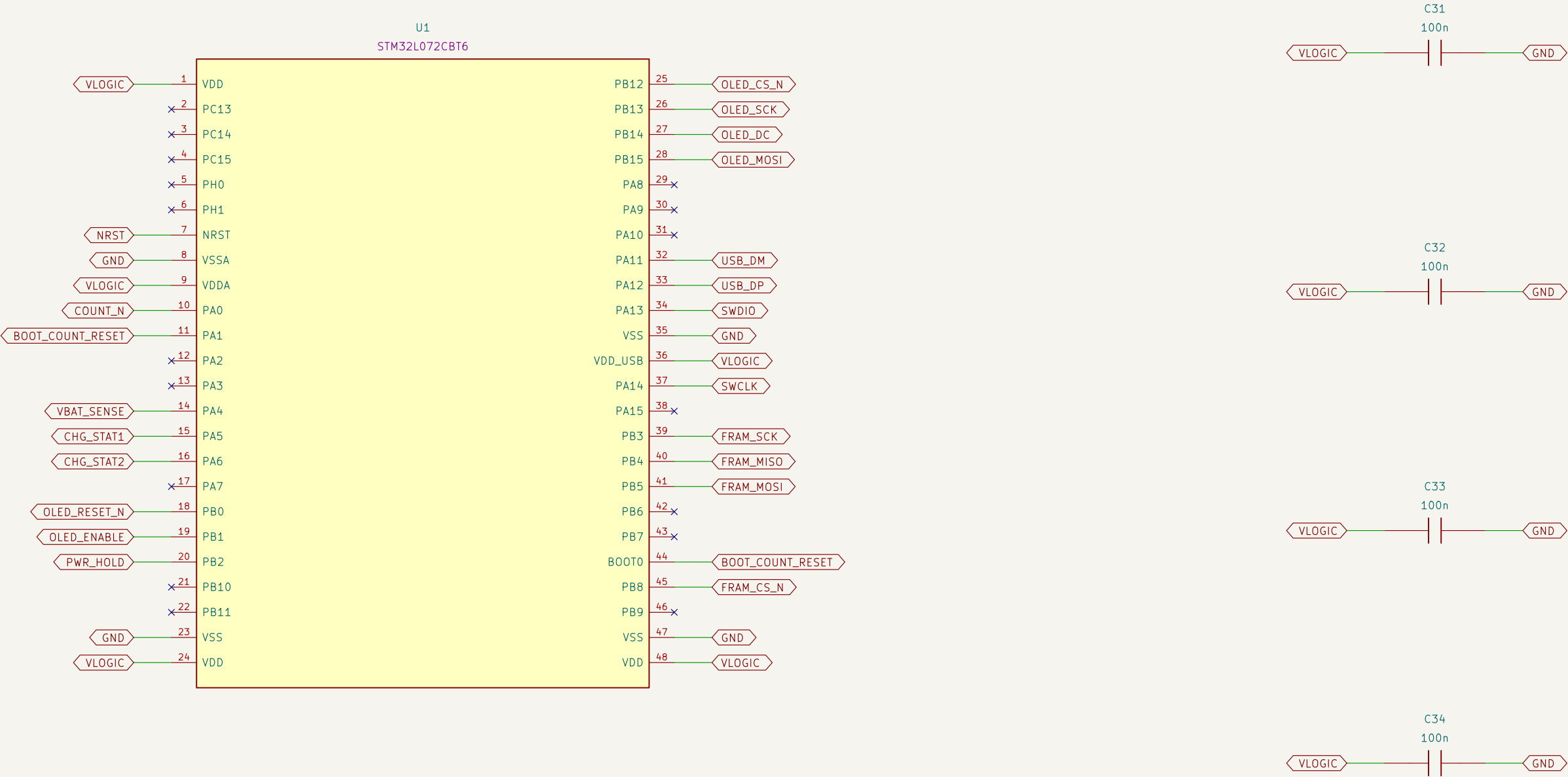
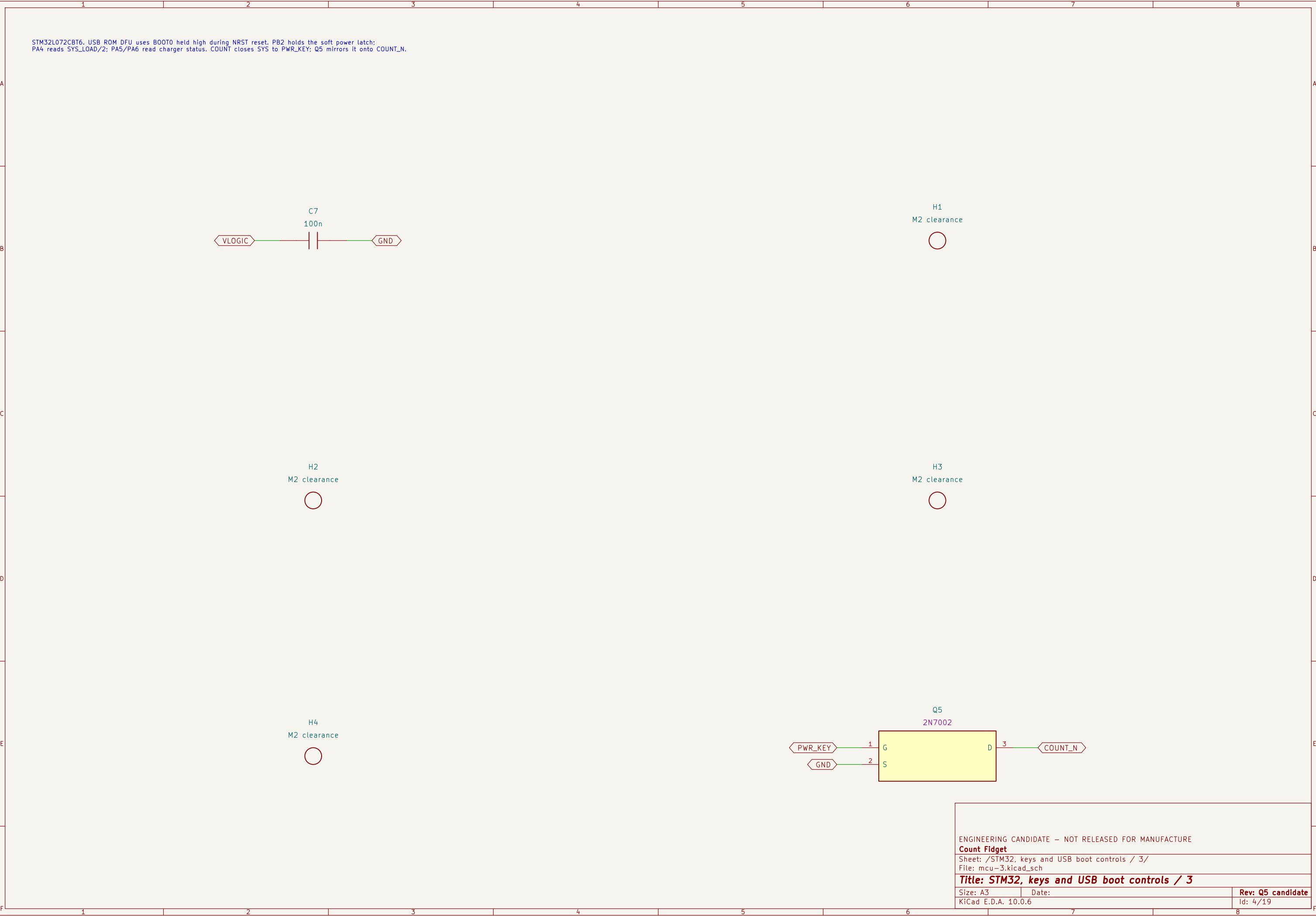


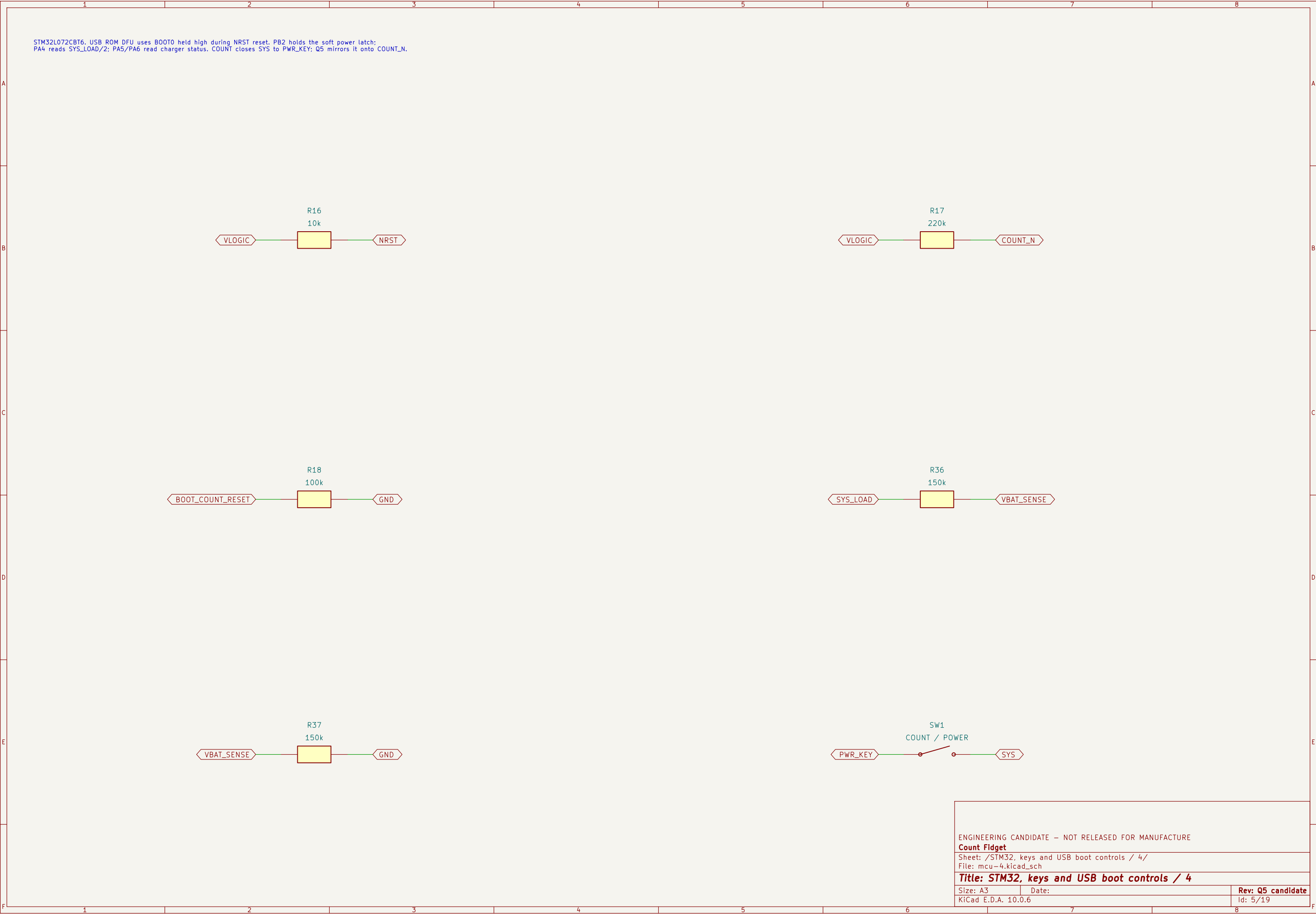
	1	2	3	4	5	6	7	8	
A	Q5 schematic index – 82 fitted PCB parts plus non–fitted test lands and mounts. Global net labels connect the functional sheets. Crosses explicitly mark unused physical pins. Q1 submitted files remain frozen. This candidate still requires ERC, layout, first–article and safety qualification.								A
	STM32, keys and USB boot controls		STM32, keys and USB boot controls / 2			STM32, keys and USB boot controls / 3			
	U1 / C31 / C32 / C33 / C34		C10 / C11 / C12 / C35 / C37 / C6			C7 / H1 / H2 / H3 / H4 / Q5			
B	File: mcu–1.kicad_sch STM32, keys and USB boot controls / 4		File: mcu–2.kicad_sch STM32, keys and USB boot controls / 5			File: mcu–3.kicad_sch STM32, keys and USB boot controls / 6			B
	R16 / R17 / R18 / R36 / R37 / SW1		SW2 / SW3 / TP1 / TP11 / TP12 / TP2			TP3 / TP4 / TP5 / TP6 / TP9			
	File: mcu–4.kicad_sch SPI OLED module and power control / 1		File: mcu–5.kicad_sch SPI OLED module and power control / 2			File: mcu–6.kicad_sch Independent SPI retained–count memory / 1			
C	C20 / C28 / D51 / R24 / R25 / R29		U6			C30 / C36 / R32 / R33 / U8			C
	File: display–1.kicad_sch USB, charging, soft power latch and common supply / 1		File: display–2.kicad_sch USB, charging, soft power latch and common supply / 2			File: memory–1.kicad_sch USB, charging, soft power latch and common supply / 3			
	C1 / C13 / C2 / C29 / C3 / C38		C39 / C40 / C5 / D1 / D2 / D3			D4 / J1 / R1 / R19 / R2 / R3			
D	File: usb–power–1.kicad_sch USB, charging, soft power latch and common supply / 4		File: usb–power–2.kicad_sch USB, charging, soft power latch and common supply / 5			File: usb–power–3.kicad_sch On–board LIR2032 holder and protection / 1			D
	R30 / R31 / R34 / R35 / R38 / R4		R5 / R6 / TP13 / U2 / U3 / U7			BT1 / C4 / Q1 / Q2 / R7 / R8			
	File: usb–power–4.kicad_sch On–board LIR2032 holder and protection / 2		File: usb–power–5.kicad_sch Independent charge–temperature window / 1			File: battery–1.kicad_sch Independent charge–temperature window / 2			
E	R9 / U4		C9 / Q3 / Q4 / R10 / R11 / R12			R13 / R14 / R15 / R39 / TH1 / U5			E
	File: battery–2.kicad_sch		File: thermal–1.kicad_sch			File: thermal–2.kicad_sch			
F						ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE Count Fidget Sheet: / File: click–counter–Q5.kicad_sch Title: Count Fidget – Q5 engineering candidate			F
						Size: A3		Date:	Rev: Q5 candidate
						KiCad E.D.A. 10.0.6		Id: 1/19	
	1	2	3	4	5	6	7	8	

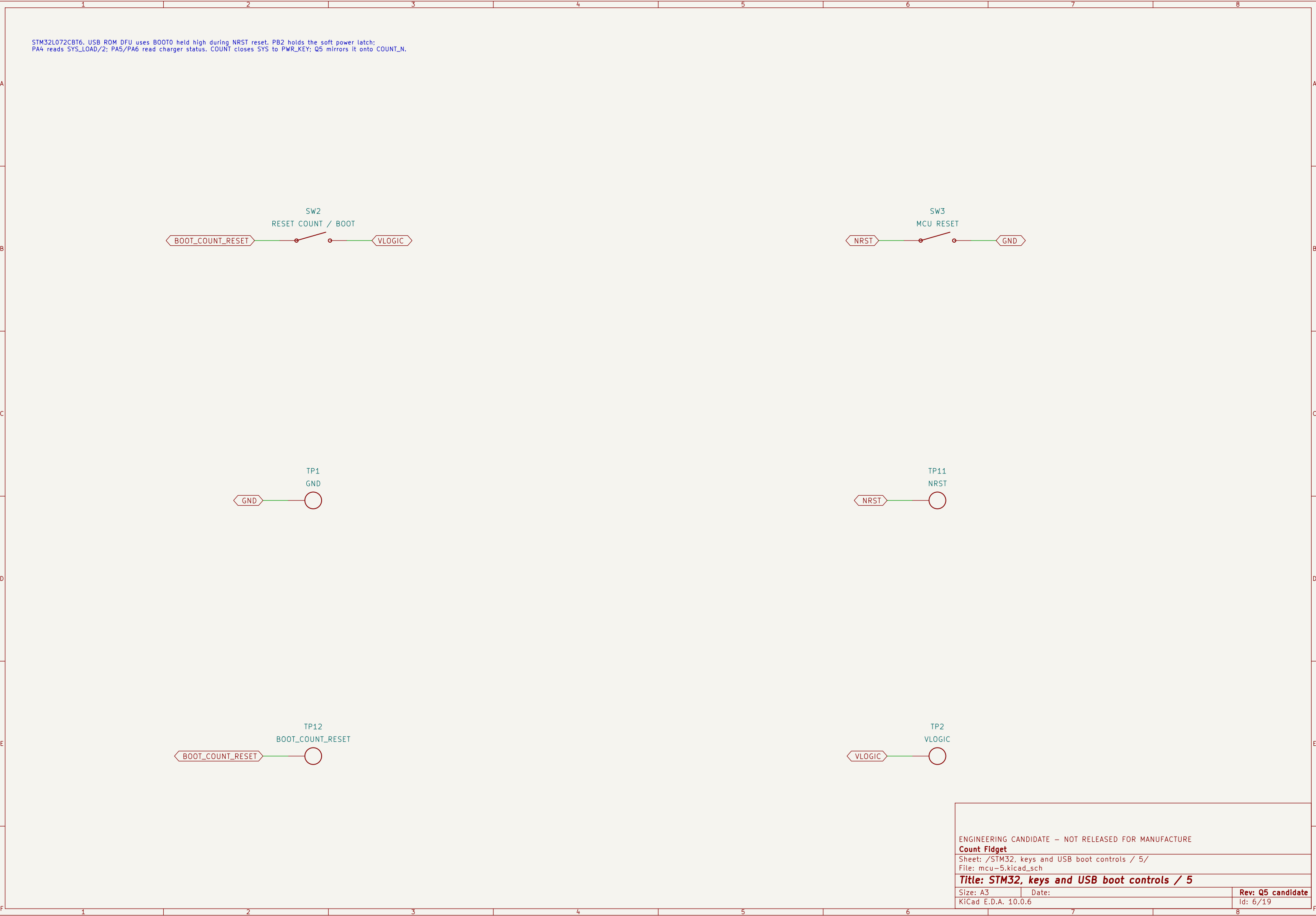
STM32L072CBT6. USB ROM DFU uses BOOT0 held high during NRST reset. PB2 holds the soft power latch; PA4 reads SYS_LOAD/2; PA5/PA6 read charger status. COUNT closes SYS to PWR_KEY; Q5 mirrors it onto COUNT_N.

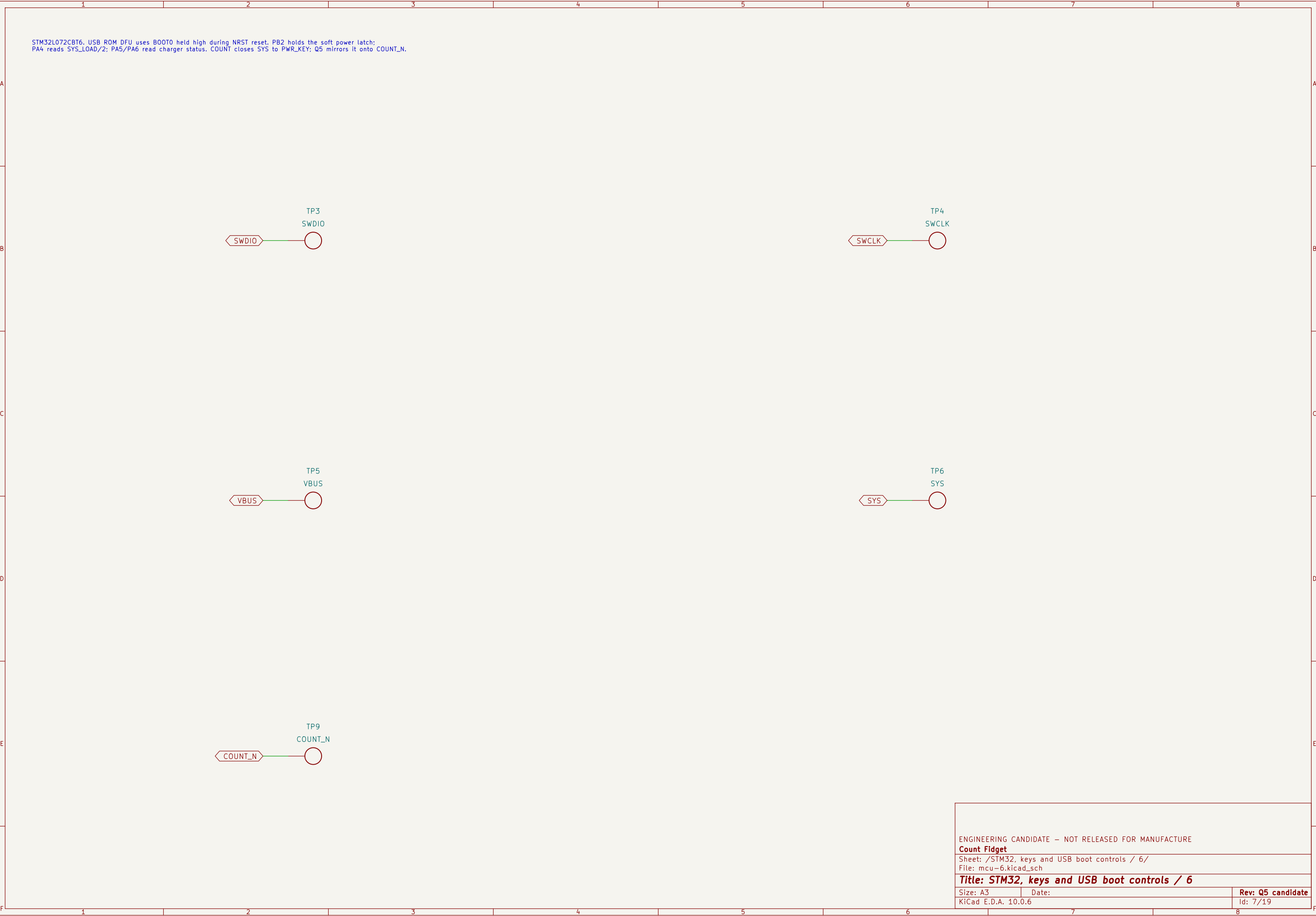


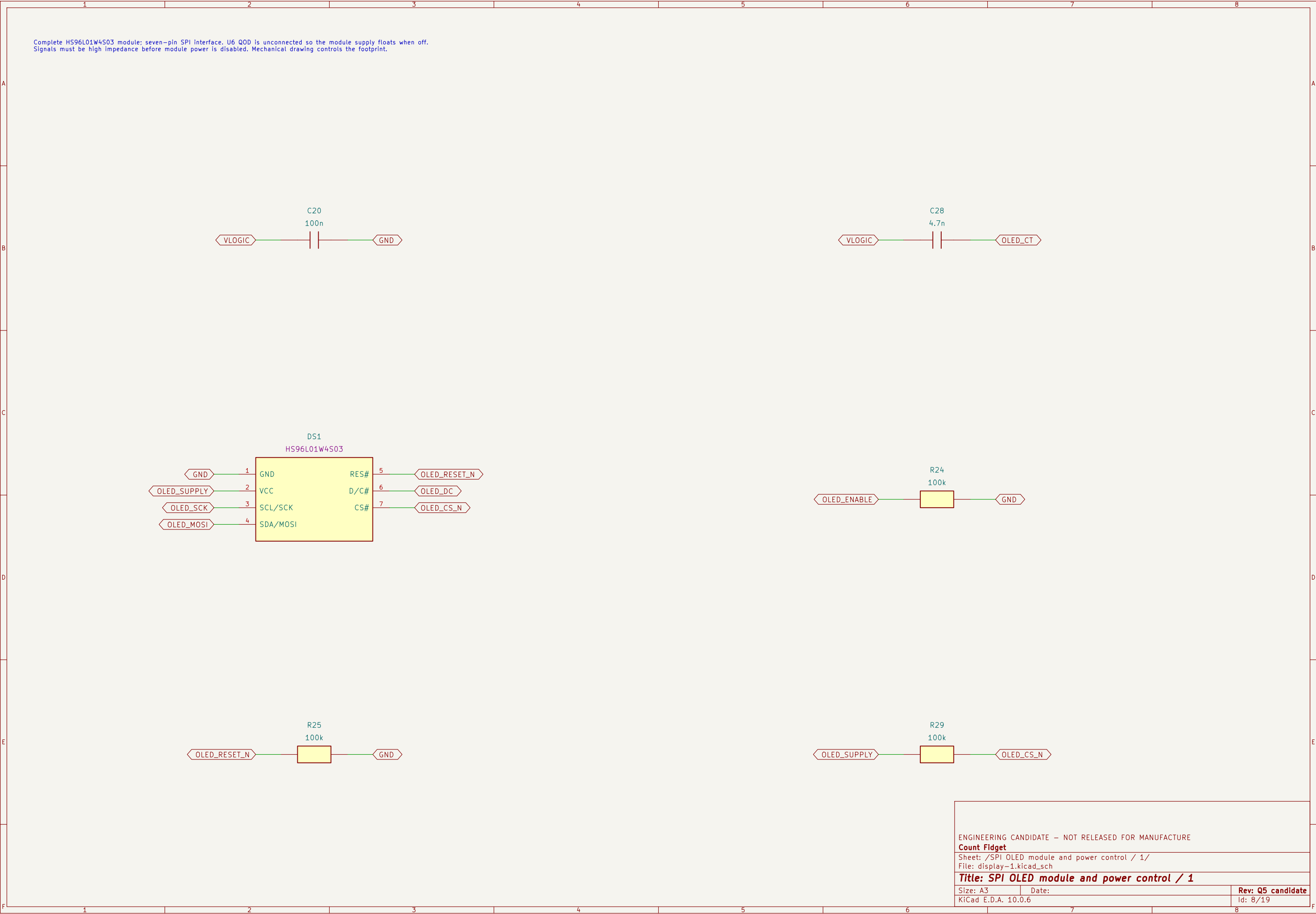


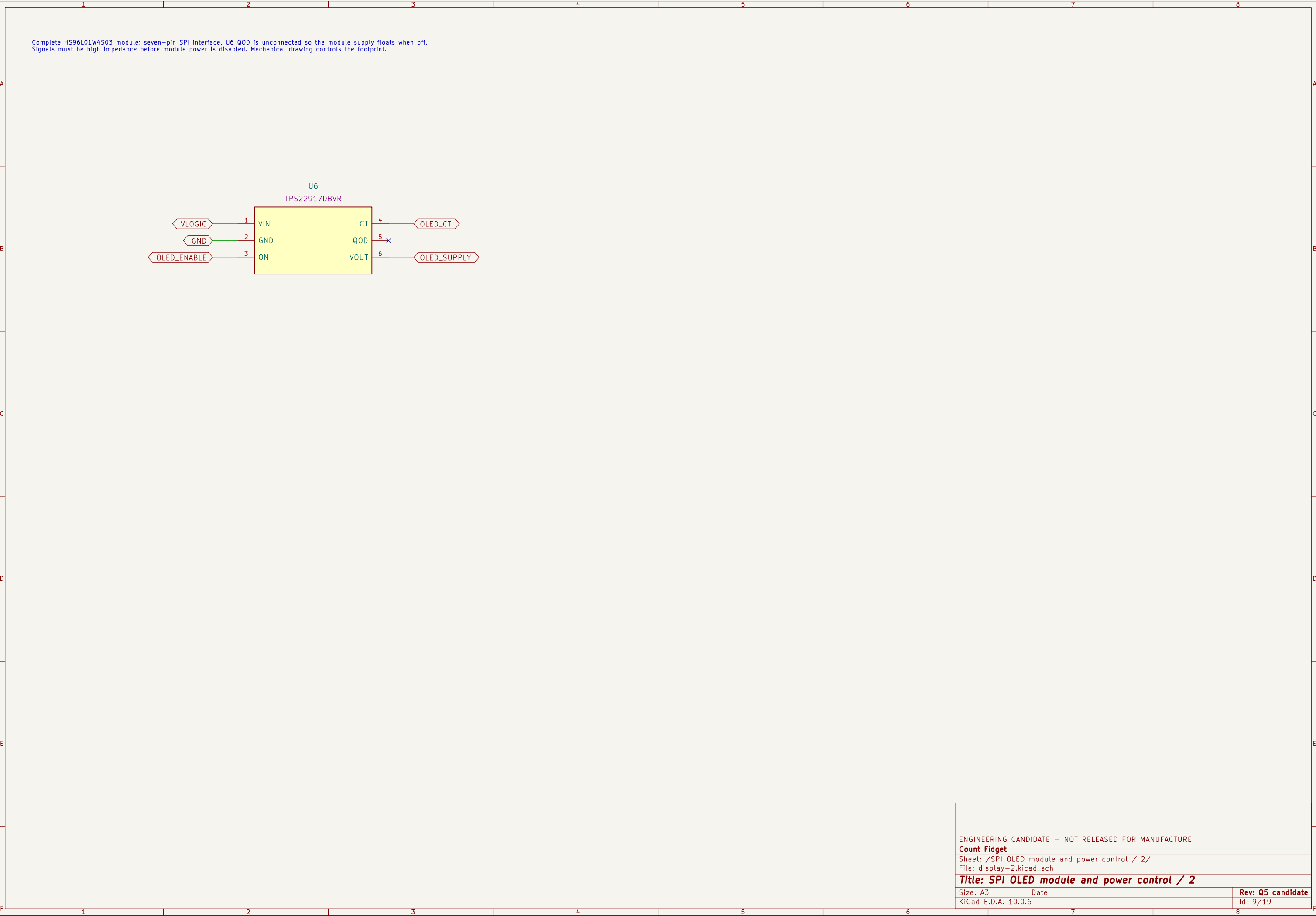


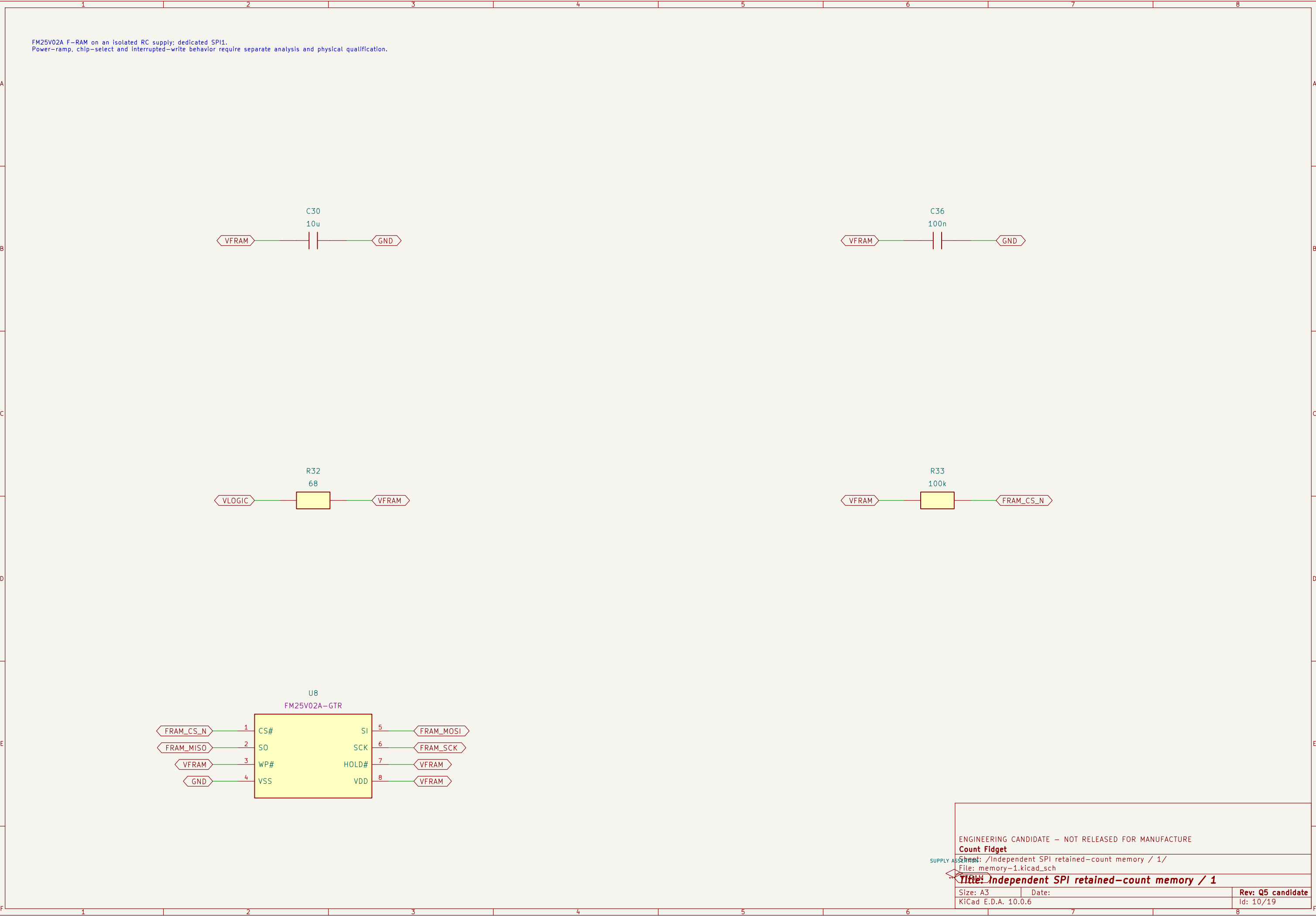












ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE

Count Fidget

Sheet: /Independent SPI retained-count memory / 1/
File: memory-1.kicad_sch

Title: **Independent SPI retained-count memory / 1**

Size: A3

Date:

KiCad E.D.A. 10.0.6

Rev: Q5 candidate

Id: 10/19

SUPPLY ASSIGNMENT

U7 ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.

C1
2.2u

C2
10u

C3
2.2u

C13
4.7n

C29
4.7n

C38
100n

SUPPLY ASSERTION
VBUS

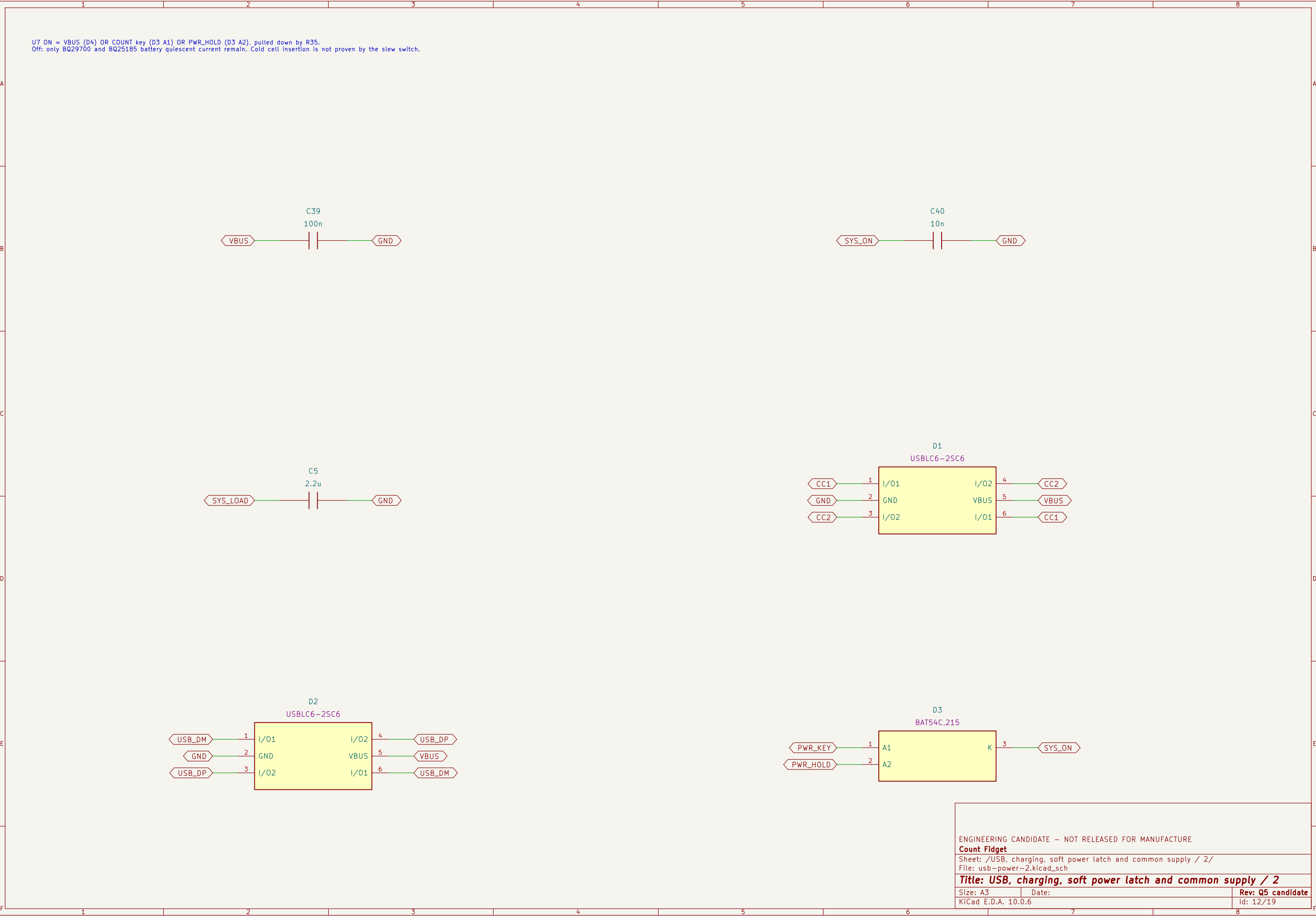
SUPPLY ASSERTION
GND

ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE
Count Fidget
Sheet: /USB, charging, soft power latch and common supply / 1/
File: usb-power-1.kicad_sch
Title: USB, charging, soft power latch and common supply / 1
Size: A3 Date: Rev: Q5 candidate
KiCad E.D.A. 10.0.6 Id: 11/19

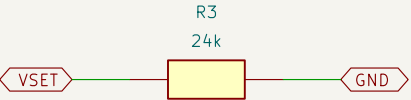
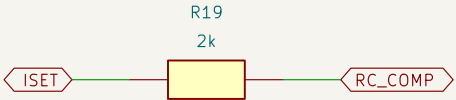
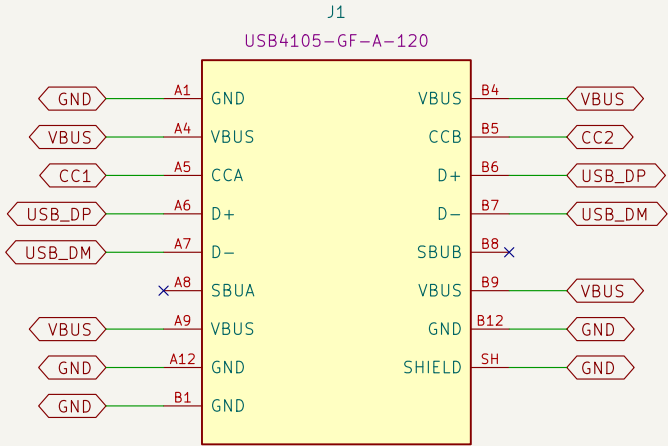
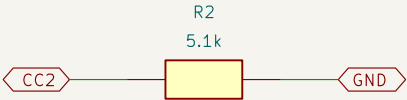
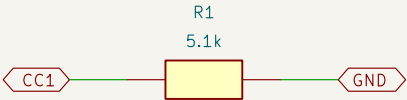
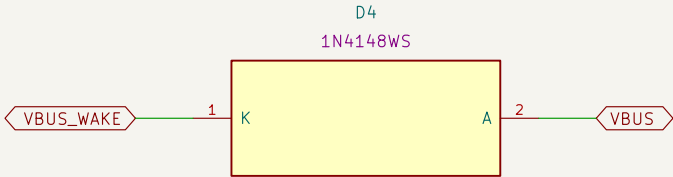
Count Fidget

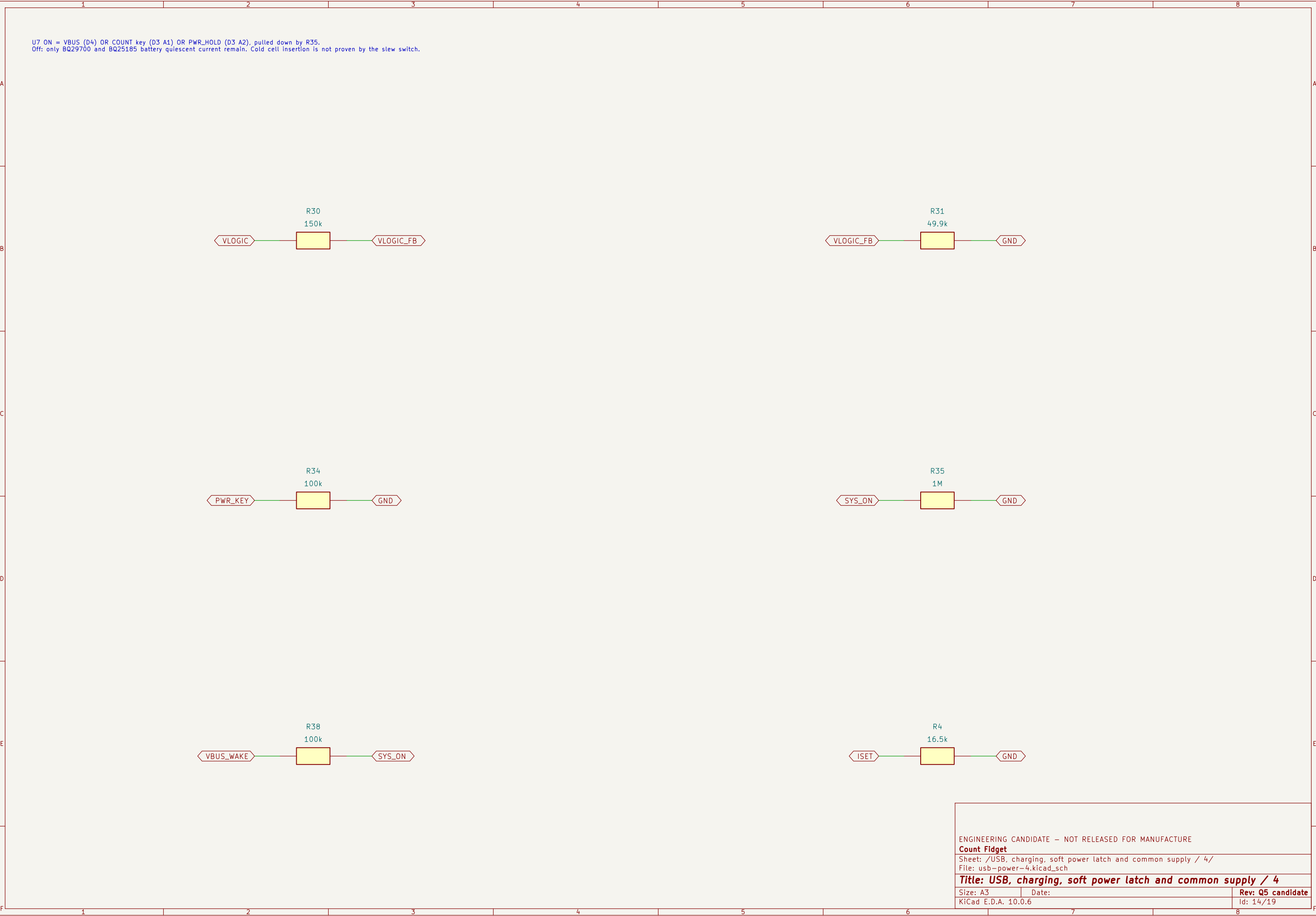
Title: USB, charging, soft power latch and common supply / 1

Size: A3	Date:	Rev: Q5 candidate
KiCad E.D.A. 10.0.6		Id: 11/19

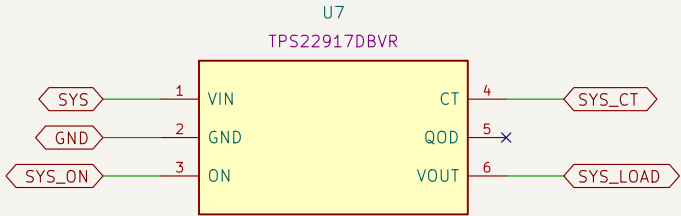
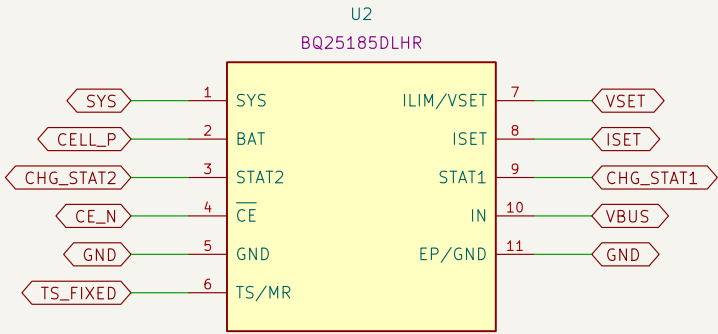
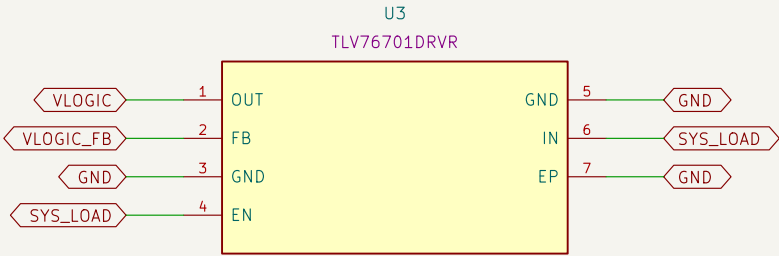
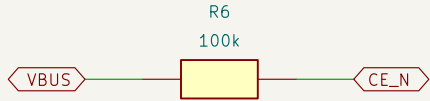
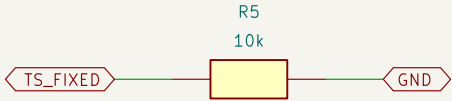


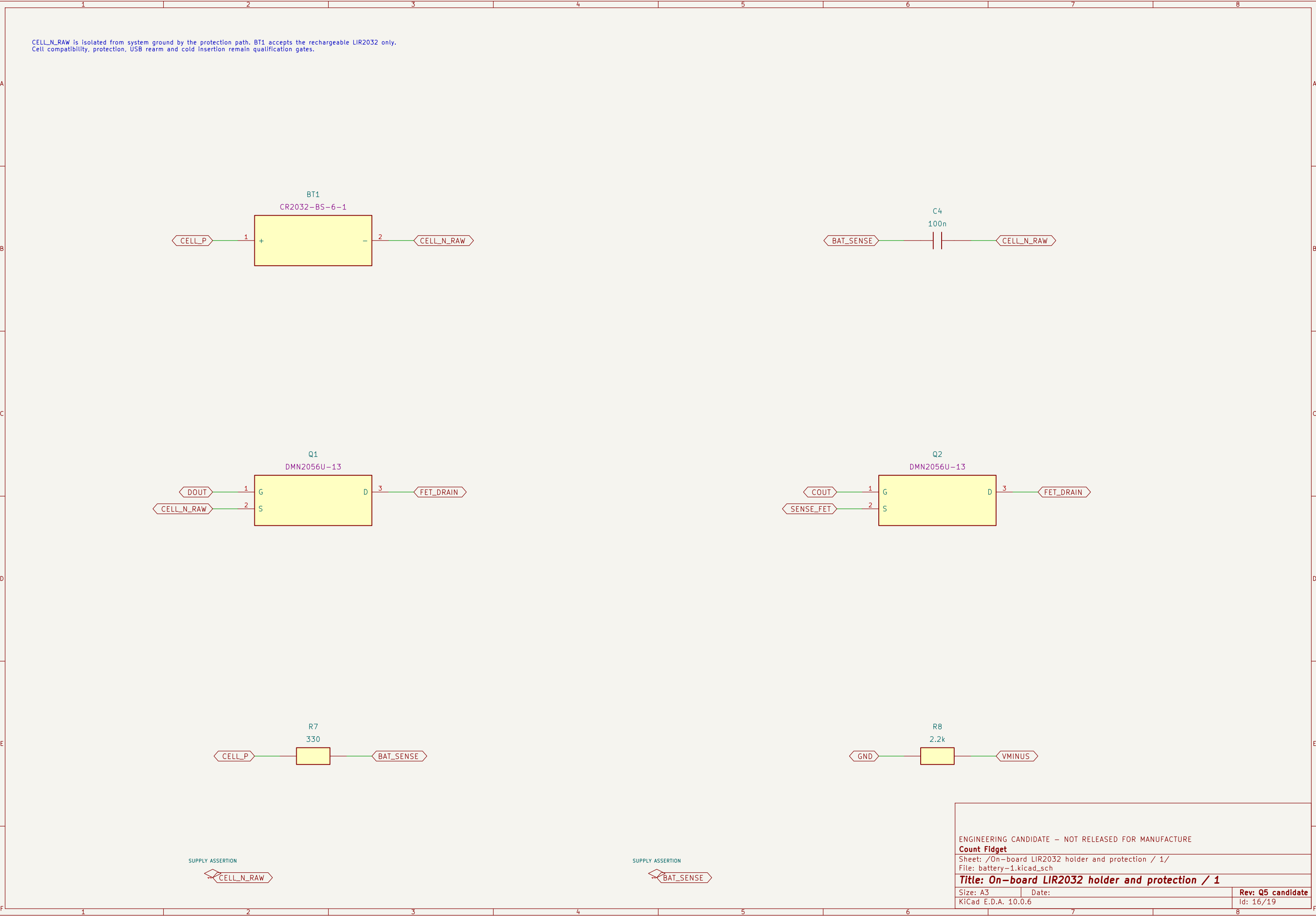
U7_ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.

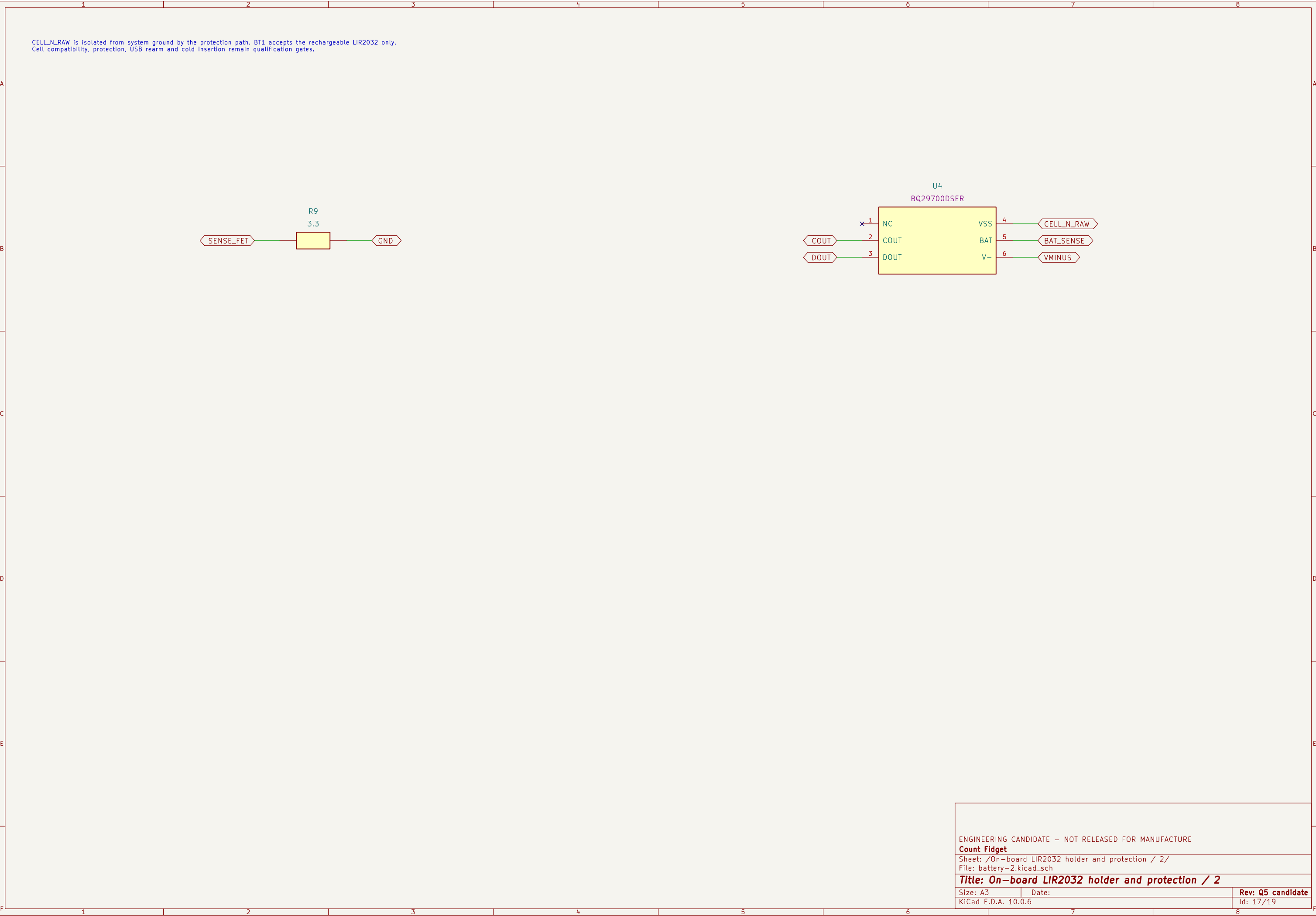




U7 ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.







TH1 on the board beside BT1; both window outputs must enable their series FETs for charging.
 Gate pulldowns disable charging when comparator power is absent; threshold and fault tests remain required.

The schematic diagram illustrates a thermal management system. It features a microcontroller (U5) with various pins connected to different components. A capacitor C9 (100n) is connected to U5_VCC and GND. Transistors Q3 and Q4 (2N7002) are used for switching; Q3's gate is connected to TEMP_COLD_OK and its source to CE_MID, while its drain is connected to CE_N. Q4's gate is connected to TEMP_HOT_OK and its source to GND, with its drain connected to CE_MID. Resistors R10 (57.6k) and R12 (24.3k) are connected between VBUS and TEMP_SENSE and TEMP_COLD/TEMP_HOT respectively. Resistor R11 (115k) is connected between VBUS and TEMP_COLD. A 5V regulator (U4) is shown with its input connected to VBUS and its output to 5V_VCC. The 5V_VCC pin is also connected to the microcontroller's VCC and the 5V pin of a connector. The microcontroller's GND is connected to the GND pin of the connector. A note at the top left specifies that TH1 on the board beside BT1; both window outputs must enable their series FETs for charging, and gate pulldowns disable charging when comparator power is absent; threshold and fault tests remain required.

ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE
 Count Fidget
 Sheet: /Independent charge-temperature window / 1/
 File: thermal-1.kicad_sch
 Title: Independent charge-temperature window / 1
 Size: A3 Date: Rev: Q5 candidate
 KiCad E.D.A. 10.0.6 Id: 18/19

